Trainings

General Information

- The use of quality engine lubricating oils, combined with appropriate oil drain and lubricating oil filter change intervals, is a critical factor in maintaining engine performance and durability.
- Cummins Inc. recommends the use of a high quality 15W-40 multi-viscosity heavy-duty engine lubricating oil that meets the requirements of Cummins® Engineering Specification (CES) 20078, or 20081 (such as Valvoline™ Premium Blue™ or Valvoline™ Premium Blue Extreme™). For areas where products meeting CES 20078, or 20081 are **not** readily available, a product meeting American Petroleum Institute (API) CH-4 or CES 20076 can be used, but at a reduced drain interval. Reference the Oil Drain Intervals by severity of service km [mi] section. The oil grades CC, CD,CE CF, CG-4, and CF-4 have been obsoleted by API and **must not** be used.
- Reference the Maintenance Schedule in the appropriate Owners or Operation and Maintenance Manuals.
- Shortened drain intervals can be required with monograde oils, as determined by close monitoring of the oil condition with scheduled oil sampling. Use of single-grade oils can affect engine oil control.
- Synthetic engine oils, API Group III and Group IV basestocks, are recommended for use in Cummins® engines operating in ambient temperature conditions consistently below -25°C [-13°F]. Above this temperature, petroleum-based multigrade lubricants are recommended. Synthetic 0W-30 oils that meet API Group III and Group IV basestocks can be used in operations where the ambient temperature **never** exceeds 0°C [32°F]. 0W-30 oils do **not** offer the same level of protection against fuel dilution as do higher multigrade oils. Higher cylinder wear can be experienced when using 0W-30 oils in high-load situations.

For further details and an explanation of engine lubricating oils for Cummins® engines, refer to Cummins® Engine Oil Recommendations, Bulletin 3810340. (/qs3/pubsys2/xml/en/bulletin/3810340.html)

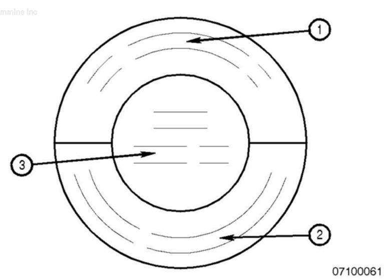
Additional information regarding lubricating oil availability throughout the world is available in the Engine Manufacturing Association (EMA) Lubricating Oils Data Book for Heavy Duty Automotive and Industrial Engines. The data book can be ordered from: Engine Manufacturers Association, Two North LaSalle Street, Suite 2200, Chicago, IL 60602; Phone: (312) 827-8700, Facsimile: (312) 827-8737 (www.enginemanufacturers.org).

Oil viscosity **must** be chosen according to the typical climate conditions experienced by the user. Use of 15W-40 is recommended for the best engine durability at higher ambient temperatures. For cold conditions 5W-30 viscosity can be used for easier starting and improved oil flow

AfterMarket Oil Additive Usage

Cummins Inc. does **not** recommend the use of aftermarket oil additives. Present high-quality fully additive engine lubricating oils are very sophisticated, with precise amounts of additives blended into the lubricating oil to meet stringent requirements defined in (1) Cummins® Engineering Specification CES 20076 that is similar to API CH-4, in (2) CES 20078 that is similar to API CI-4, and in (3) CES 20081 that is similar to API CJ-4. These furnished oils meet performance characteristics that conform to the lubricant industry standards. Aftermarket lubricating oil additives are **not** necessary to enhance engine oil performance and in some cases can reduce the finished oil's capability to protect the engine.

The API service symbols are shown in the accompanying illustration.



1. The upper half of the symbols display the appropriate oil categories.
2. The lower half contains words to describe additional oil information.
3. The center section identifies the SAE oil viscosity grade.

New Engine Break-in Oils

Special "break-in" engine lubricating oils are **not** recommended for new or rebuilt Cummins® engines. Use the same lubricating oil that will be used during normal operation.

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